Huawei Access Node HEB-ACC-40 Standard Operating Procedure for Routing Protocol Flap Resolution

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5351

Description

This Standard Operating Procedure (SOP) is designed to address the Routing Protocol Flap failure domain on Huawei Access Node HEB-ACC-40 located at Hebbal Edge. The failure domain is characterized by a high frequency of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, resulting in the generation of alarm code ALARM_5351 with a severity of Minor. The underlying cause of this failure domain lies in the inherent complexities of the routing protocol interactions, where the dynamic nature of BGP route advertisement, OSPF adjacency establishment, and ISIS LSP propagation can lead to instability and oscillations in the routing protocol state machine. Furthermore, the Huawei Access Node's configuration and operation may also contribute to the exacerbation of this issue, particularly if there are any misconfigurations or hardware-related issues affecting the node's ability to maintain stable routing protocol connections.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that remote hands are available for assistance, if required.

- Check the status of the redundant links connecting the Huawei Access Node HEB-ACC-40 to the Hebbal Edge site to ensure that they are operational and functioning as intended.

- Review the configuration of the Huawei Access Node HEB-ACC-40 to ensure that it is in compliance with the recommended best practices and guidelines for Huawei Access Nodes.

Potential Root Causes

Cause	Probability
High CPU utilization on the Huawei Access Node HEB-ACC-40 due to excessive routing protocol activity.	High
Misconfiguration of the Huawei Access Node HEB-ACC-40's routing protocol settings, leading to instability and oscillati	Medium
Hardware-related issues affecting the Huawei Access Node HEB-ACC-40's ability to maintain stable routing protocol co	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp routing-table, Expected Output: Display the current BGP routing table and identify any anomalies or inconsistencies., Description: This command will provide insight into the BGP routing table and help isolate any issues related to BGP route churn.
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Command: display ospf neighbor, Expected Output: Display the current OSPF neighbor list and identify any issues with OSPF adjacency establishment., Description: This command will provide information about the OSPF neighbor list and help identify any issues related to OSPF adjacency loss.

Command: display isis database, Expected Output: Display the current ISIS database and identify any issues with ISIS LSP propagation., Description: This command will provide information about the ISIS database and help identify any issues related to ISIS LSP flooding.
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Resolution Steps

Step 1: Step: Verify and optimize the Huawei Access Node HEB-ACC-40's routing protocol settings., Command: display current configuration | include bgp ospf isis, Description: Review the current configuration of the Huawei Access Node HEB-ACC-40 to ensure that the routing protocol settings are optimized and in compliance with the recommended best practices and guidelines.

Step 2: Step: Adjust the Huawei Access Node HEB-ACC-40's routing protocol timers and parameters to improve stability., Command: bgp timer timers bgp timers ospf timers isis, Description: Adjust the routing protocol timers and parameters to improve stability and reduce the impact of routing protocol activity on the Huawei Access Node HEB-ACC-40.

Step 3: Step: Clear the Huawei Access Node HEB-ACC-40's routing protocol connections and restart the protocols., Command: clear bgp *; reload, Description: Clear the routing protocol connections and restart the protocols to ensure that the Huawei Access Node HEB-ACC-40 is functioning correctly.

Post-Check Verification

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Command: display bgp routing-table, Expected Output: Verify that the BGP routing table is stable and free of anomalies., Description: Verify that the BGP routing table is stable and free of anomalies to ensure that the resolution steps were effective.

Command: display ospf neighbor, Expected Output: Verify that the OSPF neighbor list is stable and free of issues., Description: Verify that the OSPF neighbor list is stable and free of issues to ensure that the resolution steps were effective.

Command: display isis database, Expected Output: Verify that the ISIS database is stable and free of issues., Description: Verify that the ISIS database is stable and free of issues to ensure that the resolution steps were effective.
```

Roll-back Procedures

- Step: Restore the original configuration of the Huawei Access Node HEB-ACC-40., Command: copy current configuration to original configuration, Description: Restore the original configuration of the Huawei Access Node HEB-ACC-40 to ensure that the system is returned to its previous state in the event that the resolution steps fail.

- Step: Restart the routing protocols on the Huawei Access Node HEB-ACC-40., Command: reload, Description: Restart the routing protocols on the Huawei Access Node HEB-ACC-40 to ensure that the system is returned to its previous state in the event that the resolution steps fail.